	Branch: - Computer Science and Engineering Semester :VI Subject: - 6KS04/Big Data Analytics
	UNIT-I
1	List the three characteristics of Big Data, and explain the main considerations in processing Big Data ?
2	Define analytic sandbox and its importance?
3	Explain the Difference between BI and Data Science.
4	Describe the Challenges of the current analytical architecture for data scientist?
5	Explain Drivers of Big data, Data Evolution and rise of Big Data Sources.
6	Explain Big Data Eco system and New Approach to analytics
7	Describe exact key role for the new big data ecosystem.
8	State Data Analytics Lifecycle.
9	Explain and understand the Data Analytics Lifecycle: Discovery phase in details.
10	Explain and understand the Data Analytics Lifecycle: Data Preparation phase in details
11	Explain and understand the Data Analytics Lifecycle: Model Planning phase in details
12	Explain and understand the Data Analytics Lifecycle: Model Building phase in details
13	Explain and understand the Data Analytics Lifecycle: Communicate Results and operationalize phase in details
14	Identify the phase that team expect to invest most of the project time. Identify the team that would expect to spend the least time.
15	Illustrate various Kinds of tools to be used in the following phases, and for which kinds of use scenarios? a. Phase 2: Data Preparation a. Phase 4: Model Building
	UNIT-II
1	Define and understand Hypothesis Testing ?
2.	Define type 1 Error / Type II error? Identify the one always more serious than the others? Discuss in detail.
3.	Explain ANOVA Hypothesis Test in details
4	You are analysing two normally distributed populations, and your null hypothesis is that the mean M1 of the first population is equal to the mean M2 of the second. assume the significance level is set at 0.05. if the observed p-value is 4.33e-05, Conclude the decision regarding the null hypothesis?
5.	Describe clustering and explain different clustering methods?
6.	Explain K means Algorithms.
7	Define Association rule with Apriori algorithm in details.
8	Discuss the Evolution of candidate Rules .
9.	State the applications of association Rules
10	What is exploratory data analytics and explain its method.
11	List out motive of exploratory data analytics.
12	What is data visualization ? which are different types of Data visualizations ?
13	What is data visualization ? what are the advantages and disadvantages of Data visualizations ?
14	Explain statical methods for evaluation.
15	Differentiate between 'student's t-test' and 'welch's t-test'
16	What is hypothesis testing , Explain with ex. Null Hypothesis and alternative Hypothesis

<table>
<tr><td></td><td>UNIT-III</td></tr>
<tr><td>1</td><td>Describe how logistic Regression can be used as classifier</td></tr>
<tr><td>2</td><td>What is Regression ? Explain any one type of Regression in Detail.</td></tr>
<tr><td>3</td><td>Describe Coefficient of Regression .</td></tr>
<tr><td>4</td><td>Describe Model of Linear Regression.</td></tr>
<tr><td>5</td><td>Describe Model of Linear Regression with normally distributed errors</td></tr>
<tr><td>6</td><td>Explain importance of the categorial variables in Regression</td></tr>
<tr><td>7</td><td>Explain tools and techniques that can be used to validate a fitted linear regression model.</td></tr>
<tr><td>8</td><td>Describe , what is mean by residual standard error</td></tr>
<tr><td>9</td><td>What is N-fold cross validation Model , describe.</td></tr>
<tr><td>10</td><td>Prove that the corelation coefficient is the geometric mean between in regression coefficient that is $r^2 = b_{yx} . b_{xy}$</td></tr>
<tr><td>11</td><td>The Regression lines of a sample are x+6y=6 and 3x+2y=10 find
1. Sample means x and y
2. The coefficient of corelation between x and y
3. Also find the value of y at x=12</td></tr>
<tr><td>12</td><td>From the following results obtain the two regression equations and estimate the yield when the rainfall is 29 cm and rainfall when the yield is 600kg.
<table>
<tr><td></td><td>Yield in Kg</td><td>Rain fall in cm</td></tr>
<tr><td>Mean</td><td>508.4</td><td>26.7</td></tr>
<tr><td>SD</td><td>36.8</td><td>4.6</td></tr>
</table>
The coefficient of corelation between yield and rainfall is 0.52</td></tr>
<tr><td>13</td><td>The number of bacterial cells (y) per unit volume in a culture at different hrs. (x) is the given below
<table>
<tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr>
<tr><td>y</td><td>43</td><td>46</td><td>82</td><td>98</td><td>123</td><td>167</td><td>199</td><td>213</td><td>245</td><td>272</td></tr>
</table>
Fit lines of Regression of Y on x and X on Y, also estimate the number of bacterial cells after 15 hrs.</td></tr>
<tr><td>14</td><td>Describe Model of Logistic Regression.</td></tr>
<tr><td>15</td><td>Explain Logistic Regression , explain use cases of Logistic Regression</td></tr>
<tr><td>16</td><td>State Advantages and disadvantages of Logistic Regression.</td></tr>
</table>

17 Consider Group of 20 students spends between zero and six hrs studying an examination , how does the number of hrs. spent studying affect the probability of the student passing the exam.

1. The following table shows the number of hrs. each student spent studying and whether they passed (1) or Failed (0)

Hrs. X1	0.50	0.75	1	1.25	1.50	1.75	2	2.25	2.75	3
Pass Y	0	0	0	0	0	1	0	1	1	0

Hrs. Xi	3	3.25	3.50	4.0	4.25	4.50	4.75	5.0	5.50
Pass Yi	1	0	1	1	1	1	1	1	1

	UNIT -IV
1	Write short notes on Time Series Analysis
2	Why use autocorrelation instead of autocovariance when examining stationary time series.
3	Provide an example that if the cov(X,Y)=0 the two random variables X and Y are not necessarily independent.
4	When should an ARIMA(p,d,q) model in which d>0 be considered instead of an ARMA(p,q) model.
5	Explain the Box-jenkins Methodology of Time Series Analysis? / Explain methods of Time Series Analysis.
6	Explain ARIMA model with auto corelation function in Time Series Analysis
7.	Explain ARIMA model with auto Regreesive model and moving average models in Time Series Analysis
8.	State difference between ARIMA and ARMA model Time Series Analysis
9.	Describe steps in Text Analysis with example
10.	What are the main challenges of Text Anlysis
11	What is a corpus ? explain categories of brown corpus
12	Explain Text Analysis with ACME's process.
13	Describe Term Frequency and inverse document Frequency (TFIDF).
14	Name three benefits of using TFIDF.
15	What methods can be used for sentiment analysis
16	What is definition of topic in topic model
	UNIT-V
1.	Explain the Term Hadoop ecosystem. In details with pig, hive, HBase, mahout.
2.	Explain the task performed by map reduce.
3	Explain Pig with suitable eg.
4.	What is HBase , discuss various HBASE data model an applications.
5.	Describe the big data tools and techniques.
6.	Explain General overview of Big Data High Performance Architecture along with HDFC in details.
7.	Explain the big Data Ecosystem in details.
8.	Describe Map reduce programming Model.
9.	How do You improve the drawback of namenode.
10.	Explain, Expanding the big data application ecosystem.

11.	Compare and contrast Hadoop, Pig, Hive and HBase. List strength and weaknesses of each tool set.
12.	Differentiate Map Reduce and YARN.
13	Describe High Performance Architecture in detail.
14	What is HDFC, how it enabling storage of large files
15	Briefly Summarise HDFS working principles with suitable diagram.
16	Explain Big Data application Ecosystem
17	What are the variables to be consider when framing big data environment.
18	Explain five V's of Big Data
19	Explain Execution Model which effectively manages the workload.
	UNIT-VI
1	What is NOSQL?
2	Explain Schema less models which increasing flexibility for data manipulation?
3	Explain Key-Value Store in NoSQL.
4	Differentiate Key-Value Store and Document Store.
5.	Describe Tabular store in terms of to manage structured data
6.	Describe Object Data Store in terms of Schema less management.
7.	Explain in brief Graph Database
8	What is Graph analytics.
9	Explain in details simplicity of graph model
10	Explain how graph analytics is applied in enhancing health care quality.
11	List and describe in details application areas of graph analytics.
12	Explain how graph analytics is applied in cyber security.
13	Explain Graph analytics algorithm and solution approaches.
14	What are technical complexities arises to analyse graphs.
15	What are the features of graph analytics platform, explain in details
16.	Explain Data visualization basics in terms of graph analytics.